

EX 9:

Compare Linear and Polynomial Regression using Python

PROGRAM:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.preprocessing import PolynomialFeatures
```

```
from sklearn.metrics import mean_squared_error
```

```
# Dataset
```

```
X = np.array([1, 2, 3, 4, 5, 6]).reshape(-1, 1)
```

```
y = np.array([2, 5, 10, 17, 26, 37])
```

```
# Linear Regression
```

```
linear = LinearRegression()
```

```
linear.fit(X, y)
```

```
linear_pred = linear.predict(X)
```

```
# Polynomial Regression
```

```
poly = PolynomialFeatures(degree=2)
```

```
X_poly = poly.fit_transform(X)
```

```
polynomial = LinearRegression()
```

```
polynomial.fit(X_poly, y)
```

```
poly_pred = polynomial.predict(X_poly)
```

```
# Errors
```

```
print("Linear Regression MSE:",  
      mean_squared_error(y, linear_pred))
```

```
print("Polynomial Regression MSE:",  
      mean_squared_error(y, poly_pred))
```

```
# Graph
```

```
plt.scatter(X, y, label="Actual Data")  
plt.plot(X, linear_pred, label="Linear Regression")  
plt.plot(X, poly_pred, label="Polynomial Regression")
```

```
plt.xlabel("X")  
plt.ylabel("Y")  
plt.title("Linear vs Polynomial Regression")  
plt.legend()  
plt.show()
```

OUTPUT

